



OPEN Fault tolerant cluster-based routing in wireless sensor network using novel intelligent deep learning methodology

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Wireless Sensor Networks (WSNs) are highly susceptible to cluster head (CH) failures due to limited node energy and unstable communication links, resulting in data loss and reduced network reliability. This work presents a fault-tolerant cluster-based routing model that integrates an Improved Graph Neural Network (IGNN) with Multi-Objective Crested Porcupines Optimization (MOCPO) for efficient recovery from CH failures. The model constructs a virtual CH at the sink node by organizing the residual energy and connectivity information of failure-free CHs. The proposed IGNN uses essential parameters such as node position, Euclidean distance, residual energy, connectivity degree, packet delay, and fault-tolerance score to identify optimal fault-recovery paths. MOCPO optimizes IGNN parameters under a multi-objective fitness function incorporating energy consumption, end-to-end delay, connectivity measure, and fault-tolerance metrics. Simulation results with 1000 sensor nodes, 10 faulty CHs, packet size of 500 bytes, and 200 optimization iterations show that the proposed IGNN-MOCPO model reduces energy consumption by 32.72% and end-to-end delay by 33.14%, while significantly improving packet delivery ratio, residual energy, and network lifetime compared to existing FPCR schemes. These results demonstrate the efficiency, stability, and adaptability of the proposed framework under dense and fault-prone WSN deployments.

Keywords Wireless sensor network, Fault tolerant cluster-based routing, Enhanced graph neural network, Multi-objective crested porcupines optimization

List of symbols

$G(V, E)$	Graph representation of the WSN, where V is the set of sensor nodes and E is the set of communication links
V	Set of all sensor nodes in the network
E	Set of communication edges between nodes
N	Total number of sensor nodes in the network
i, j	Indices representing sensor nodes
(x_i, y_i)	Coordinates of node i in the 2D deployment space
d_{ij}	Euclidean distance between node i and node j
R	Transmission range of a sensor node
l	Length of the data packet (in bits)
E_{tx}	Energy consumed for transmitting data
E_{rx}	Energy consumed for receiving data
E_{elec}	Energy dissipated per bit in transmitter/receiver electronics
ϵ_{fs}	Amplifier energy for free-space model
ϵ_{mp}	Amplifier energy for multipath fading model
d_0	Threshold distance separating free-space and multipath models

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E_{DA}	Energy required for data aggregation per bit
E_{total}	Total communication energy consumed in the network
p	Desired percentage of cluster heads (CHs)
$T(n)$	Threshold function for CH selection
r	Current round number
G'	Set of nodes not selected as CHs in recent rounds
f	Fitness function used in optimization
w_1, w_2, w_3, w_4, w_5	Weight coefficients for multi-objective fitness components
F_t	Normalized fault tolerance score
E_r	Residual energy of cluster heads
C_m	Connectivity measure
D_{avg}	Average end-to-end delay
E_c	Energy consumption metric
D_{max}	Maximum allowable delay threshold
h	Number of hops in routing path
N_c	Number of connected nodes
N_t	Total number of nodes
E_{init}	Initial energy of sensor nodes
$E_{res,i}$	Residual energy of node i
N_{CH}	Number of cluster heads
N_{nbr}	Number of neighboring nodes
$N_{backup}^{(k)}$	Number of backup CHs or alternate paths
$x_i^{(k)}$	Feature vector of node i at iteration k in IGNN
$\mathcal{N}(i)$	Set of neighboring nodes of node i
$W^{(k)}$	Weight matrix at IGNN layer k
$\sigma(\cdot)$	Activation function (ReLU)
Z	Graph-level embedding output
P	Population size in MOCPO
T	Number of iterations in MOCPO
D	Dimensionality of solution vector
X_i	Position (solution vector) of the i^{th} porcupine
$f_m(X)$	Objective function $\min$ multi-objective optimization
$g(X), h(X)$	Inequality and equality constraints
CD	Crowding distance in Pareto optimization
X_{best}	Best solution obtained so far
α, β	Control parameters in MOCPO update equations

WSNs have emerged as a crucial methodology for many real-world uses, such as healthcare, smart agriculture, industrial automation, military surveillance, as well as environmental monitoring¹. In order to detect, analyse, as well as transmit information to a central BS or sink, these frameworks are composed of several tiny, low-energy sensor nodes dispersed over a geographic area². Energy economy, scalability, as well as reliable data transmission are critical concerns in WSNs because of their impromptu and often widespread deployment³. Cluster-based routing has been widely used as a crucial strategy to address these problems. Communication burden is reduced as well as energy consumption is optimized by clustering sensor nodes under the supervision of a CH⁴. To increase the network's lifetime, the CH gathers data from member nodes and sends it to the BS. However, CHs are essential, and their failure can lead to data loss as well as network fragmentation or drastically reduce network effectiveness⁵.

The vulnerability of cluster-oriented routing protocols to mistakes as well as malfunctions describes a major issue with WSNs⁶. Sensor nodes, like CHs, can malfunction due to communication connection failures, hardware problems, harsh environmental conditions, or energy depletion⁷. The entire nodes in the cluster may lose communication with the rest of the framework if a CH fails, which might lead to increased loss of information, energy waste from retransmissions, as well as even a network failure⁸. Fault management is a difficult task as these errors can occur suddenly as well as often without notice. Network resilience is decreased by traditional cluster-oriented routing protocols like LEACH as well as HEED since they don't have enough ways to handle these failures on the fly⁹. In order to increase the reliability as well as strength of information transmission in WSNs, fault tolerance combination into cluster-oriented routing protocols has been a significant study field in recent years¹⁰.

To lessen the detrimental effects of node as well as CH failures, many fault-tolerant techniques have been proposed¹¹. These include strategies like backup CHs, multi-path routing, re-clustering, and dynamic CH substitution. These strategies often result in increased energy costs, increased computing complexity, or communication delays, even if they can improve fault tolerance to a certain extent¹². Furthermore, ensuring that fault-tolerant techniques can successfully adjust to the real-time status of the network, such as mobility, changing node energy levels, or topological variations, remains a difficulty¹³. Long-term WSN operation depends on intelligent, flexible, as well as energy-efficient fault-tolerant frameworks¹⁴. In order to operate within the limited resources of typical sensor nodes, a successful clustering-based fault-tolerant routing system must balance network connection, minimize energy consumption, as well as guarantee speedy failure recovery¹⁵. This necessitates researching sophisticated algorithms as well as architectural layouts that may deliver consistent effectiveness in spite of unforeseen errors. The general structure of WSN is shown in Fig. 1.

The paper contribution is as below.

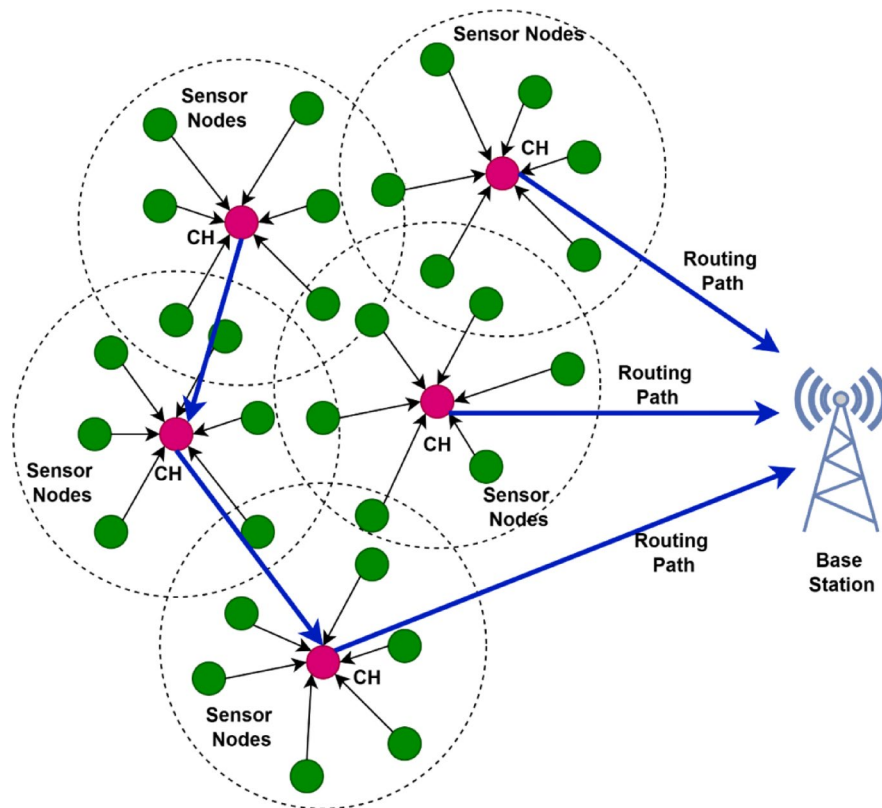


Fig. 1. General structure of WSN.

- To perform the FTCR in WSN using novel intelligent deep learning-oriented optimization framework.
- To efficiently handle CH failures using deep learning and virtual CH construction.
- To act as a common backup for entire malfunctioning CHs by carefully organizing the resources available from entire failure-free CHs as a virtual CH.
- To achieve fault tolerance and reduce the total energy consumption of entire failure-free CHs using novel IGNN, where the parameter tuning is done by the MOCPO algorithm.

The paper organization is as follows. Section 1 is the introduction of FTCR in WSN model. Section 2 is literature survey. Section 3 is proposed methodology with proposed model, network model, energy consumption model, fitness function, virtual CH, novel IGNN for FTCR and MOCPO algorithm. Section 4 is results and analysis. Section 5 is the conclusion.

Motivation

Strong, fault-tolerant communication systems are urgently needed, as seen by the increasing reliance on WSNs for critical tasks including combat surveillance, environmental monitoring, as well as catastrophe detection. In particular, cluster-oriented routing protocols are very susceptible to the failures of CHs, which serve as the primary communication nodes for their clusters, while being energy-efficient as well as scalable. Data collection as well as transfer can be disrupted by a single CH failure, which lowers network dependability and causes the loss of important information. This vulnerability is exacerbated in harsh or dynamic conditions in which node failures frequently happen as a result of energy depletion, device deterioration, or connection issues. The inability of traditional routing protocols to handle these problems promptly often results in frequent re-clustering, significant overhead, as well as a shorter network lifespan. These challenges underline the need for robust approaches that provide a steady as well as efficient information flow having minimal energy consumption, in addition to quickly detecting and addressing faults. Fulfilling this condition can significantly increase WSNs' operating consistency, lifetime, as well as dependability. Therefore, developing a smart, adaptable, as well as energy-efficient FTCR architecture is crucial to meeting the growing demands of modern WSN applications used in unpredictable as well as resource-constrained environments.

Related work

To tackle the fault-tolerant issue in WSNs with a hierarchical topology, a hybrid fault-tolerant routing technique was proposed¹⁶. Clustering and the assignment of sensor nodes as Gaussian integers were combined in the hierarchical format. Consequently, the network area was divided into tiny square grids, where the CH of every grid was represented by a Gaussian integer. By connecting these CHs, a Gaussian network was created. The hybrid Fault-tolerant Clustering routing protocol for Wireless sensor networks (FCGW) was presented in this

research. It took use of node symmetry, multi-path routing's advantages, as well as the Gaussian network's shortest distances. FCGW aimed to improve data dependability, reduce energy consumption, as well as increase fault tolerance in WSNs.

The goal of the study was to examine the many types of faults that could occur in a WSN¹⁷. Enhancing sensor networks' resilience as well as capacity for self-healing was the primary objective of the fault-tolerant approach. Numerous failure kinds in WSN were frequently caused by the remote deployment, resource-constrained environment, as well as harsh monitoring circumstances. There were two approaches to solving the problem: fault tolerance or defect prevention. The novel aspect of this state-of-the-art study was the authors' introduction of two novel algorithms: Population-oriented Clustering, which focused on fault avoidance, as well as Uniform Energy Clustering, which aimed at fault tolerance.

To tackle the problem of fault tolerance in hierarchical topology WSNs, a dynamic fault-tolerant routing framework was designed^{18,19}. The hierarchical topology also included clustering as well as the labelling related to sensor nodes as Gaussian integers. As a result, the region associated with the framework was separated into smaller square grids, where an integer based on the Gaussian distribution represented the CH of every grid. All of these CHs have been connected to form a Gaussian network. On the basis of the Gaussian framework, this research suggested a hybrid fault-tolerant clustering routing strategy for WSNs. Node symmetry, the Gaussian network's minimum distance, and the benefits of multi-path routing (FCGW) were all used in this protocol.

On the basis of failures in WSNs, a novel fault-tolerant routing technique was presented to address the difficulties in hierarchical topology²⁰. Clustering and labelling sensor nodes as Gaussian integers were integrated in hierarchical topology. Consequently, the network area was separated into tiny square grids. The CH created an integer with a Gaussian distribution to describe every grid. A Gaussian network was produced by these node clusters. This study also investigated the benefits of multi-path routing as well as introduced a novel fault-tolerant routing technique by utilizing node symmetry using WSNs with a Gaussian network. This research's primary objectives were to increase data throughput, provide dependable fault tolerance, as well as advance energy effectiveness.

To efficiently handle CH failures, flow graph modelling as well as virtual CH generation were used²¹. To act as a common backup for entire CHs that malfunction, the resources from entire failure-free CHs were first methodically organized as a virtual CH. The flow graph modelling technique was then used to provide fault tolerance while reducing the overall energy consumption of entire operational CHs. Finally, extensive experiments were carried out to demonstrate the effectiveness of the approach in Internet of Things (IoT) WSN fault-tolerant routing.

Using the Self-Clustering Routing (SCR) as well as Self-Clustering Routing with Dynamic Trust and Recovery Aware (SCR-DTRA) algorithms, this endeavor proposed a self-repairing technique for fault-tolerant data aggregation²². While SCR-DTRA increased fault tolerance using trust-oriented node selection as well as adaptive recovery approaches, SCR enhanced energy use through dynamic clustering. By improving error recovery, data accuracy, as well as network lifetime, these techniques made WSNs more robust for real-time applications. Even in the event of failures, the proposed method ensured continuous information flow. Results from simulations showed significant improvements in network stability and energy efficiency over traditional techniques. This approach increased communication reliability and reduced data duplication. Applications that require accurate as well as continuous information tracking would particularly benefit from it. Some of the features and challenges of the considered conventional works are listed in Table 1 below.

Derakhshanfard et al.³⁰ proposed an SVM-based abnormal behavior detection framework for information-centric IoT networks to address the challenges of inaccurate detection, changing traffic conditions, and limited computational resources. The method uses network characteristics such as packet reception rate, delay, and energy consumption to classify node behavior as normal or abnormal. Evaluation using the MDC and UNSW-NB15 datasets demonstrated improved accuracy and F1-score compared with conventional SVR and denoising-based SVM approaches, while maintaining stable performance under increased delay and packet loads. The study therefore provides an efficient machine-learning solution for improving IoT network reliability, security, and energy-aware operation.

Li et al.³¹ developed a low-voltage and low-power integrated RF transmitter specifically designed for 2.4-GHz wireless sensor network applications. The proposed architecture combines an LC voltage-controlled oscillator with a sub-harmonic mixer to achieve frequency stability, low phase noise, reduced power consumption, and lower harmonic distortion. Implemented using GlobalFoundries 0.18-μm CMOS RF technology, the transmitter operates from a 1-V supply while consuming only 2.55 mW. The design also achieves an IIP3 of 3.19 dBm and approximately 12.5-dB conversion gain, demonstrating its suitability for energy-constrained WSN nodes requiring reliable and efficient wireless communication.

Citation	Methodology used	Features	Challenges
²³	EEFRP	Improves energy efficiency	It depends mostly on appropriate parameter estimation
²⁴	FTEER	Enhanced network lifetime	Scalability issues in vast deployments
²⁵	EFT-PMD	Enhances data availability	Needs important computation for real-time applications
²⁶	Weighted multi-objective function	Bypasses void areas	Energy cost is more
²⁷	Flower pollination algorithm	Robust against CH faults	Complex backup
^{28,29}	i-FTEECR	Minimizes the effect of CH failure	Limits optimal route selection scheme

Table 1. Features and challenges of the conventional works considered.

Xiao et al.³², investigated the problem of reliable narrowband interference detection in wireless communication systems, particularly under complex channel conditions and dynamically varying interference environments. Their work exploits unlabeled data to enhance the robustness of interference detection, addressing the limitations of conventional approaches that depend heavily on labeled training samples. By incorporating additional unlabeled information into the detection process, the proposed approach aims to maintain reliable identification performance when propagation conditions and interference characteristics change. This study demonstrates the potential of semi-supervised or data-efficient learning strategies for improving interference awareness and communication reliability in practical wireless networks.

Xiong et al.³³, proposed a low-cost physical-layer security framework for intelligent reflecting surface (IRS)-assisted massive MIMO systems using one-bit digital-to-analog converters. The study addresses the high hardware complexity and cost associated with high-resolution DACs in large-scale MIMO systems and numerous IRS elements. The proposed design jointly optimizes one-bit quantized transmit precoding and constant-modulus IRS phase shifts to maximize the secrecy rate. By reducing RF hardware requirements while retaining secure transmission capabilities, the approach provides an attractive solution for practical large-scale wireless systems where security, energy efficiency, and implementation cost must be considered simultaneously.

Qi et al.³⁴ introduced RVH-Cover, a resultant-vector- and heap-based optimization algorithm for improving coverage performance in large-scale wireless sensor networks. The work addresses the fundamental challenge of balancing sensing coverage quality with computational efficiency as the number of deployed sensor nodes increases. The algorithm uses resultant-vector-based processing together with heap-based optimization to efficiently determine sensor deployment or coverage configurations. This approach is particularly relevant to IoT and WSN applications where maximizing monitored areas while limiting computational overhead and improving resource utilization are essential. The study therefore contributes an efficient optimization strategy for scalable and effective sensor-network deployment.

Wang et al.³⁵, proposed GEMS, a GAT-embedded mobility framework for managing dual connectivity in integrated terrestrial and non-terrestrial networks. The study addresses the difficulty of maintaining reliable secondary satellite links when terrestrial and satellite coverage continuously changes because of satellite mobility, heterogeneous link characteristics, and limited coverage windows. GEMS formulates dual-connectivity allocation as a bipartite graph-matching problem and uses a graph attention network to generate user-specific mobility-aware link weights. The resulting approach jointly considers instantaneous throughput and long-term service continuity, achieving improved service rates and signaling efficiency compared with conventional repeated and snapshot-based strategies.

Collectively, these studies demonstrate that next-generation IoT and wireless networks increasingly rely on **machine learning, hardware-efficient architectures, intelligent optimization, physical-layer security, and mobility-aware networking** to overcome limitations in reliability, energy consumption, interference management, security, coverage, and connectivity. These complementary approaches provide important foundations for developing scalable and intelligent communication architectures for future 5G/6G, WSN, IoT, and integrated terrestrial–non-terrestrial networks.

Problem statement

Due to sensor nodes' limited energy, computation, as well as communication capabilities, WSNs are prone to faults by nature. Cluster-oriented routing describes the most popular of the many routing techniques because of its energy efficiency and scalability. However, CHs, who are responsible for BS communication and data aggregation, are crucial to this approach. A single CH failure brought on by hardware malfunctions, energy exhaustion, or unfavorable climatic conditions can cut off a whole node cluster, causing network fragmentation and data loss. It is common for existing cluster-oriented routing protocols to lack effective mechanisms for detecting as well as quickly resolving these errors. As a result, in problematic circumstances, they are unable to maintain dependable effectiveness and network connectivity. Furthermore, a lot of the fault-tolerant techniques now in use need costly fixed backup or re-clustering frameworks, which raise energy consumption as well as delay communication. The primary problem describes the absence of an adaptable, intelligent, as well as energy-efficient framework that can guarantee quick recovery, seamless fault diagnosis, as well as even real-time resource allocation. Therefore, addressing the issue of reliable as well as adaptive fault-tolerant routing remains a major and unsolved task in the field of WSNs.

Proposed methodology

Novelty and distinctiveness of the proposed IGNN-MOCPO framework

The proposed IGNN-MOCPO architecture introduces several original contributions that extend beyond merely combining existing techniques. First, the framework develops a virtual cluster head (CH) abstraction, where the residual energy and connectivity resources of all operational CHs are logically organized at the sink node to support multiple faulty clusters. This dynamic backup mechanism is not used in earlier GNN-based routing studies and enables rapid recovery from CH failures without re-clustering. Second, the proposed model incorporates an Improved Graph Neural Network (IGNN) that embeds fault-related attributes—node energy, spatial coordinates, connectivity degree, and delay estimates—directly into the graph structure, allowing the learning model to adapt its routing decisions to varying levels of fault severity and node density. Third, a Multi-Objective Crested Porcupines Optimization (MOCPO) strategy is introduced specifically to tune the IGNN parameters under simultaneous objectives of reducing energy consumption, minimizing delay, enhancing connectivity, and improving fault-tolerance. Finally, the framework establishes a joint IGNN–MOCPO learning–optimization cycle that enables continuous adaptation to network changes without increasing communication burden on sensor nodes. These components collectively differentiate the IGNN-MOCPO model from prior

GNN-driven routing methods and form a unique, intelligent, and fault-adaptive solution for maintaining robust communication in WSN environments.

Proposed model

A new fault-tolerant routing technique for WSNs is provided in this section. This method combines two approaches: deep learning and virtual CH. The sink node may determine the failure event when specific CHs fail. The sink node may also learn the remaining energy levels associated with every CH by routinely receiving sensed information from the CH. Furthermore, the sink node is not affected by energy and memory constraints. Using the collected available energy information, the sink node may logically generate a virtual CH by carrying out intricate operations. The sink node first verifies the maximum fault-tolerant capacity offered by entire failure-free CHs using the virtual CH. A deep learning method known as innovative IGNN is then used by the sink node to capture entire fault-tolerant possibilities among functioning as well as faulty CHs. A nature-oriented optimization method used to improve the parameters in GNN describes the MOCPO algorithm. The optimal fault-tolerant transmission pairings among malfunctioning as well as functional CHs are determined by the sink node. In order for the failure-free CHs to take over control of the respective defective CHs, the sink node sends the fault-tolerant information to them. The proposed FTCT in WSN model is diagrammatically shown in Fig. 2.

Network model

In this network, a two-dimensional plane $Y \times Z$ larger than $E \times E$ has o sensor nodes, O_j ($o = 0, 1, 2, 3 \dots o-1$), dispersed at random. If the Euclidean distance ($e_{j,k}$) between two nodes O_j and O_k is less than or equal to the transmission range (E), then they can communicate directly. Assume the coordinates of nodes O_j and O_k be represented as (y_j, z_j) and (y_k, z_k) , respectively. Therefore, using Eq. (1), the Euclidean distance ($e_{j,k}$) among O_j and O_k is determined.

$$e_{j,k} = \sqrt{(y_k - y_j)^2 + (z_k - z_j)^2} \quad (1)$$

Two nodes are called neighbor nodes if they have direct communication capabilities. As shown in Fig. 3, nodes that are not directly linked to the Base Station (BS) node choose one among their nearby nodes to serve as the forwarding node in order to send their sensed information to the BS.

Energy consumption model

For transmitting an l -bit packet over distance d , the radio energy dissipation model is:

$$E_{TX}(l, d) = \begin{cases} lE_{elec} + l\epsilon_{fs}d^2, & d < d_0 \\ lE_{elec} + l\epsilon_{mp}d^4, & d \geq d_0 \end{cases} \quad (2)$$

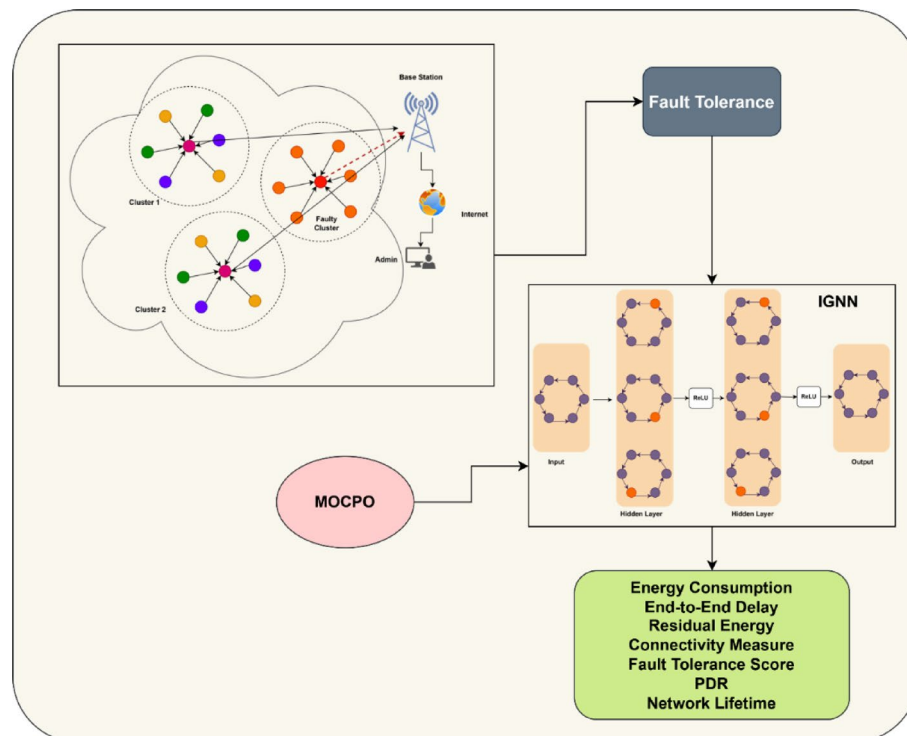


Fig. 2. Proposed FTCT in WSN model.

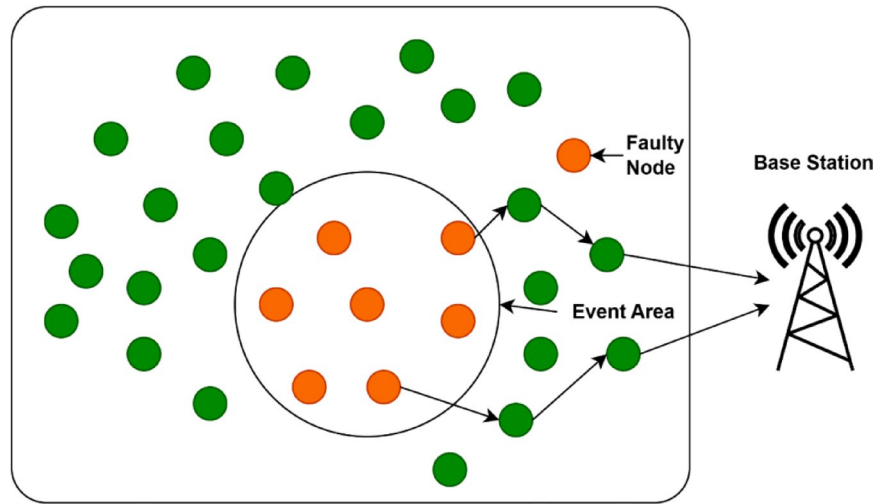


Fig. 3. Multi-hop transmission from event area to sink.

Receiving energy:

$$E_{RX}(l) = lE_{elec} \quad (3)$$

Total communication energy for a CH or node:

$$E_{comm} = E_{TX}(l, d) + E_{RX}(l) \quad (4)$$

Energy per bit of data aggregation:

$$E_{DA} = lE_{da} \quad (5)$$

Total energy consumption over all CHs.

$$E_{total} = \sum_{i=1}^{N_{CH}} (E_{TX}^{(i)} + E_{RX}^{(i)} + E_{DA}^{(i)}) \quad (6)$$

A sensor node is labelled as CH for the present rotation round only when the random count selected by SN falls beneath the threshold as shown in Eq. (2).

$$U(j) = \begin{cases} \frac{pch}{1 - pch \times \left(\text{round} \left(\frac{1}{q} \right) \right)} & \text{if } j \in A \\ 0 & \text{else} \end{cases} \quad (7)$$

Here, A describes the set of nodes that have not functioned as CH in the preceding $\frac{1}{pch}$ rounds, $round$ denotes the present round count, and pch describes the intended CH proportion in WSNs. Equation (3) illustrates the radio energy dissipation method utilized in WSN to transmit a message of ‘ n ’ bits across a distance ‘ e ’.

$$FU_y(n, e) = \begin{cases} n * F_{Elec} + n * \varphi_{gt} * e^2, & e < U_j \\ n * F_{Elec} + n * \varphi_{nq} * e^4, & e \geq U_j \end{cases} \quad (8)$$

The ratios of φ_{gt} to φ_{nq} describes constants that show the amplifier energy needed to maintain a respectable Signal-to-Noise Ratio (SNR), in which F_{Elec} stands for electronic energy. The open space channel method (e^2 power loss) is employed when the distance is less than the threshold value U_j , and the multi-path fading channel method (e^4 power loss) is utilized when the distance is more than U_j . The receiver goes through similar procedures. Equation (4) displays the energy employed for radio.

$$F_{Sy}(n) = n * F_{Elec} \quad (9)$$

Compared to the energy consumed for communication, less energy is utilized for calculation as well as storage. Therefore, just the energy consumed for communication—which is calculated as displayed in Eq. (5) is considered for simplicity.

$$F_{Ey} = t * l * F_{Merge} \quad (10)$$

The energy required to transmit or receive a 1-bit message is represented by the F_{Elec} of Eqs. (3) and (4). Equation (5)'s F_{Merge} indicates the energy required to combine a 1-bit message.

Fitness function

Enhancing CH selection as well as route development while ensuring energy efficiency, fault tolerance, and network longevity describes the aim. By lowering latency as well as energy consumption while improving connection, fault tolerance, and remaining energy, the major goal is to select the optimal CHs and routing paths. Given the following definition of a solution G 's fitness:

$$G = \omega_1 \cdot F_{Norm} + \omega_2 \cdot E_{Norm} + \omega_3 \cdot D_{Norm} + \omega_4 \cdot S_{Norm} + \omega_5 \cdot G_{Norm} \quad (11)$$

Here, the weights are described by ω_j such that $\sum \omega_j = 1$ and it reflects the priority associated with every measure, normalized fault tolerance score to be maximized is shown by G_{Norm} , normalized residual energy of CHs to be maximized is shown by S_{Norm} , normalized connectivity measure to be maximized is shown by D_{Norm} , normalized end-to-end delay to be minimized is shown by E_{Norm} and normalized energy consumption to be minimized is shown by F_{Norm} respectively. The end-to-end delay is described as below.

$$E_{Norm} = \frac{E_{Current}}{E_{Max}} \quad (12)$$

Here, the maximum tolerable delay threshold is shown by E_{Max} and the present average delay related to the packet delivery is shown by $E_{Current}$ respectively. The connectivity measure is described as below.

Delay is normalized to a threshold requirement:

$$D_{norm} = 1 - \frac{D_{avg}}{D_{th}} \quad (13)$$

Where:

- D_{avg} = measured average packet delay.
- D_{th} = maximum tolerance delay.

Expanded delay for an m -hop path:

$$D_{avg} = \sum_{h=1}^m (D_{queue}^{(h)} + D_{trans}^{(h)} + D_{prop}^{(h)}) \quad (14)$$

Thus, the effective delay metric used in the fitness function:

$$F_{delay} = D_{norm} \quad (15)$$

Here, the total count of nodes is shown by O_{Total} and the count of nodes having a path to the CH or sink is shown by $O_{Connected}$ respectively. The residual energy associated with CHs is shown as follows.

$$S_{Norm} = \frac{1}{O_{ch}} \sum_{j=1}^{O_{ch}} \frac{F_j}{F_{Init}} \quad (16)$$

Here, the initial energy per node is shown by F_{Init} , residual energy related to the j^{th} CH is shown by F_j and the count of CHs is shown by O_{ch} respectively. The fault tolerance score is described as follows.

$$G_{Norm} = \frac{1}{O_{ch}} \sum_{j=1}^{O_{ch}} \left(\frac{O_{Backup}(j)}{O_{Neighbors}(j)} \right) \quad (17)$$

Here, the count of direct neighbors related to CH j is shown by $O_{Neighbors}(j)$ and the count of backup CHs or alternate paths for CH j is shown by $O_{Backup}(j)$ respectively.

Fault tolerance considers neighbor support and backup CH availability:

$$FT = \frac{Deg(i) + B(i)}{Deg_{max} + B_{max}} \quad (18)$$

Where:

- $Deg(i)$ = number of direct neighbors connected to CH i .
- $B(i)$ = number of backup CHs or alternate forwarding paths.
- Denominator normalizes the score between 0 and 1.

If applied over all CHs:

$$FT_{score} = \frac{1}{N_{CH}} \sum_{i=1}^{N_{CH}} FT(i) \quad (19)$$

Virtual CH

This approach starts by logically creating a virtual CH on the sink node while accounting for the overall energy available from entire CHs that haven't failed. The sensed data from the sensor nodes that were first supported by operational CHs (the sensed data from non-failure members) should also be relayed by the virtual CH in addition to the members affected by failure. To represent all the data discovered from the failure-free as well as failure-affected members in the virtual CH, a virtual super frame arrangement is also used. Note that the virtual CH is logically constructed from error-free CHs and is not a physical object.

In order to increase the network's energy efficiency as well as fault tolerance, the concept of a virtual CH is suggested for WSNs. A CH is responsible for gathering information from its cluster members and transmitting it to the BS in traditional cluster-oriented routing protocols. However, communication interruption as well as significant losses of information can occur when a CH collapses. The Virtual CH method proposes adding a virtual equivalent to the real CH in order to address this issue. This virtual CH maintains an updated version of important parameters including positions, node IDs, as well as energy levels associated with the cluster members while simulating the operations and condition of the real CH. Without having to rebuild the complete cluster, the virtual CH seamlessly takes over whenever a CH dies because of a node or connection failure. By avoiding frequent re-clustering, this redundancy saves the network energy as well as ensures that data transport is interrupted as little as possible. The network's lifespan is increased and load balancing is further aided by the dynamic selection of the virtual CH, which considers proximity to the real CH and residual energy. The Virtual CH approach provides a robust way to provide reliable communication as well as fault tolerance in WSNs.

Novel IGNN for FTCT

The IGNN used in this study is intentionally designed as a lightweight graph-based learning module so that it can operate efficiently at the sink node without adding complexity to sensor nodes. The architecture consists of an input layer representing node attributes (residual energy, spatial coordinates, connectivity degree, and local delay estimate), followed by two graph convolution layers that perform neighborhood-aware feature aggregation. A ReLU activation function is used after each convolution to introduce non-linearity and stabilize information propagation across the graph. Node-level embeddings are updated using a sum-based aggregation mechanism, which enables efficient message passing without increasing computational burden. Training is performed at the sink using supervised optimization guided by the multi-objective fitness function, and parameter tuning is executed through MOCPO rather than back-propagation-heavy deep learning. The key hyperparameters include learning rate, number of convolution iterations, aggregation radius, and hidden dimension size, all optimized through the MOCPO search process. As a result, the IGNN remains computationally efficient while enabling accurate, adaptive fault-tolerant routing decisions.

The novel IGNN is used to achieve fault tolerance and reduce the total energy consumption of entire failure-free CHs. Here, the parameter tuning in GNN is done by the MOCPO algorithm. By effectively leveraging the inherent graph structure related to cluster-oriented WSNs, GNNs offer significant advantages for fault-tolerant routing. With sensor nodes acting as vertices and communication lines acting as edges, WSNs may be efficiently shown as graphs. GNNs are adept at drawing conclusions from graph-structured information, which helps with error recovery as well as routing decision-making. Their ability to comprehend complex topological patterns as well as spatial linkages describes a major advantage, allowing for flexible clustering and routing techniques that take energy levels, connectivity, and node failures into account. GNNs are perfect for quickly varying environments because, unlike traditional heuristic or rule-oriented methods, they can adaptively adjust their understanding related to the network when new information (like energy depletion or failures) emerges. Additionally, they may generalize to other network topologies, improving scalability as well as reducing the need for reprogramming. Furthermore, GNNs provide distributed inference as well as learning, which minimizes communication overhead and aligns with the decentralized nature of WSNs. GNNs increase network lifespan, energy efficiency, as well as dependability in the event of a failure by comprehending optimal routing paths and robust cluster configurations. As a result, adding GNNs to FTCT provides a reliable, data-driven approach to managing WSN complexity as well as improving network performance in general.

Despite their potential, GNNs have a number of disadvantages when used to FTCT in WSNs. One among the main problems is the high computational complexity as well as resource demands associated with inference and training, which often conflict with the limited processing power, memory, and energy resources of typical sensor nodes. Using GNN methods in a distributed as well as energy-constrained WSN environment can lead to rapid energy consumption as well as increased latency, which compromises the goal of increasing network lifetime. Furthermore, in real-time WSN implementations where nodes may operate in harsh or difficult-to-reach environments, it might be difficult to secure the vast amounts of training information as well as the appropriately labelled dataset that GNNs require. The completeness as well as quality associated with the input graph information are crucial for GNN performance; inaccurate or missing data from node failures or shaky connections can seriously reduce method accuracy. Furthermore, the complexity of creating, refining, as well as training GNN methods can result in significant development overhead and may call for specific knowledge that is frequently lacking in embedded or sensor network application domains. It is difficult to retain GNNs' real-time flexibility in rapidly shifting network topologies since it is computationally expensive as well as usually impracticable to retrain or modify methods on the fly in WSN scenarios.

By addressing a number of issues with traditional GNNs and enhancing effectiveness as well as reliability, an inventive IGNN technique offers substantial advantages for FTCT in WSNs. Through the incorporation of enhancements such as localized training, edge-aware messaging, as well as lightweight structures, the IGNN can perform better in the resource-constrained environment of WSNs. Through adaptive learning of the network topology as well as low-cost defect identification, these improvements enable the method to execute real-time, energy-efficient routing and clustering. By rerouting data away from malfunctioning nodes as well as dynamically

adjusting clusters, the strategy significantly increases the robustness and endurance of networks. Furthermore, the IGNN may integrate node-level characteristics such as link quality, communication cost, as well as residual energy, enabling context-sensitive and energy-effective decisions. The IGNN is more suited for dynamic as well as unpredictable WSN settings by reducing its need on centralized administration and large datasets through the use of decentralized inference and partial training techniques. Additionally, it enhances scalability, making it capable of handling larger networks with diverse node distributions. In contrast to conventional machine learning or heuristic approaches, a novel IGNN offers a robust, intelligent, as well as resource-adaptive solution for FTCT in WSNs, offering increased energy economy, faster adaptation, and better accuracy.

With W representing the group of sensor node vertices $W = \{w_1, w_2, \dots, w_O\}$ and F representing the group of communication link edges $F = \{f_{jk} | \text{sensor node } j \text{ is linked to sensor node } k\}$, the sensor network is represented mathematically as graph $H = (W, F)$. The nodes' properties are determined once the network is represented as a graph. Features i_j , which are described by the formula below, indicate the position as well as energy of each node w_j .

$$i_j = (q_j, F_j) \quad (20)$$

Here, F_j indicates the sensor node's remaining energy and q_j indicates the sensor node's location (y_j, z_j) . The nodes' energy as well as location are determined by their characteristics, and they update their statuses by collecting information from neighboring nodes. The below describes the expression for the node update rule for iteration $l + 1$.

$$i_j^{(l+1)} = \sigma \left(X_1 i_j^{(l)} + \sum_{k \in O(j)} X_2 i_k^{(l)} \right) \quad (21)$$

Here, σ describes a nonlinear activation function, X_1 and X_2 are weight matrices, and $O(j)$ describes the set of node (j) 's neighbors. The entire network setup may be described as below after several rounds of message transmission, yielding a graph-level output.

$$i_H = S \left(\{i_j^{(L)} | j \in W\} \right) \quad (22)$$

Here, S stands for a function that creates a graph-level description by combining node properties. Whereas i_j denotes the node j 's characteristics, i_H describes the output at the graph level.

Detailed architecture of the improved graph neural network (IGNN)

The proposed IGNN is designed as a lightweight and fault-aware graph learning framework tailored for resource-constrained WSN environments. Unlike conventional GNN models, which primarily focus on structural learning, the IGNN explicitly incorporates network reliability factors into the learning process.

Each sensor node is represented as a graph vertex with a multi-dimensional feature vector comprising residual energy, spatial coordinates, connectivity degree, and estimated delay. These features enable the model to capture both topological and operational characteristics of the network.

The IGNN architecture consists of two graph convolution layers. In each layer, node representations are updated through neighborhood aggregation using a sum-based message passing scheme. This approach reduces computational overhead compared to attention-based or dense aggregation methods. A nonlinear activation function (ReLU) is applied after each aggregation step to enhance feature expressiveness while maintaining stability.

The key distinction of the IGNN lies in its fault-aware feature embedding and adaptive learning strategy. Unlike traditional GNNs that rely on static node features, the IGNN dynamically updates node representations based on real-time network conditions, including node failures and energy variations. Furthermore, parameter tuning is not performed using conventional backpropagation-heavy training but is instead optimized using the MOCPO algorithm. This hybrid learning–optimization strategy reduces computational complexity while ensuring convergence toward optimal routing decisions.

In addition, the IGNN employs localized message propagation, limiting information exchange to neighboring nodes within a defined radius. This design choice minimizes communication overhead and aligns with the decentralized nature of WSNs.

Overall, the IGNN improves upon conventional GNN models by introducing fault-awareness, reducing computational burden, enhancing adaptability, and enabling efficient integration with multi-objective optimization.

MOCPO algorithm

By systematically improving routing decisions as well as cluster configuration to maximize network effectiveness, optimization techniques are essential for increasing FTCT in WSNs. Because sensor nodes have limited energy, computational, as well as communication capabilities, optimization techniques help ensure reliable data transmission while maximizing resource utilization. In fault tolerance, optimization makes it easier to respond to node or link failures by dynamically reassigning rerouting data, CHs as well as balancing load throughout the system. These algorithms may simultaneously optimize many goals, like determining the optimal backup nodes to maintain connectivity after failures or selecting the optimal CH on the basis of distance, remaining energy, as well as connectivity. Additionally, optimization contributes to a balanced allocation of energy across nodes, preventing the early depletion of critical nodes as well as extending the network's overall lifespan. By integrating optimization into fault-tolerant routing, WSNs may maintain data integrity, adapt in real-time to

changing conditions, as well as guarantee continuous operation in critical applications such as environmental monitoring, industrial automation, and disaster response. The MOCPO algorithm is selected here for tuning the parameters of the GNN model with the aim of deriving the fitness function.

MOCPO uses a crowding distance as well as sorting strategy that is elite and non-dominated. MOCPO is based on the recently released Crested Porcupines Algorithm as well as draws inspiration from the predator-prey dynamics related to the crested porcupines. MOCPO is made to deal with competing objectives in multi-objective optimization problems. MOCPO improves solution variety as well as convergence to the Pareto front using non-dominated sorting and crowding distance strategies. To better handle convergence as well as variety, MOCPO makes use of a new Information Feedback Mechanism (IFM) and an enhanced solution update approach.

A variant of the CPO designed specially to handle multi-objective optimization problems is the MOCPO. In order to balance exploration as well as exploitation while tackling optimization difficulties, the algorithm mimics the defensive behaviors of crested porcupines, involving scent, hearing, sight, as well as physical defence. Equation (23) may be used to characterize the multi-objective optimization problem.

$$\min_y G(y) = [g_1(y), g_2(y), \dots, g_n(y)] \quad (23)$$

It is within the restrictions listed in Eq. (15).

$$h_j(y) \leq 0, j = 1, 2, \dots, q, i_k(y) = 0, k = 1, 2, \dots, r \quad (24)$$

Here, the inequality and equality constraints are shown by $h_j(y)$ and $i_k(y)$, fitness functions are shown by $g_l(y)$ ($l = 1, 2, \dots, n$) and the design vector in e -dimensional space is shown by $y = [y_1, y_2, \dots, y_e]$ respectively.

As seen in Eq. (16), each crested porcupine represents a possible solution.

$$Q = \{y_1, y_2, \dots, y_o\} \quad (25)$$

Here, y_j is the j^{th} solution vector and o is the population size. Equation (26) provides an example of fitness evaluation.

$$G(y_j) = [g_1(y_j), g_2(y_j), \dots, g_n(y_j)] \quad (26)$$

Assign o porcupines to the following places randomly: $y_j^k \in [y_{Min}^k, y_{Max}^k]$, $\forall j, k$, in which k is the variable index and y_{Min}^k, y_{Max}^k are the bounds. The cyclic population reduction strategy dynamically adjusts the population size in accordance with Eq. (27) to enhance convergence as well as preserve variety.

$$O = O_{Min} + (O' - O_{Min}) \times \left(1 - \frac{u \bmod U}{U}\right) \quad (27)$$

Here, O : The current population. O' : The initial size of the population. O_{Min} : The smallest population size that is acceptable. u : Display the version. U : Period of repetition. Exploration has to do with the defence mechanisms of hearing as well as vision. According to Eq. (28), Vision mimics the process of examining distant regions, encouraging in-depth investigation.

$$y_j^{u+1} = y_j^u + \alpha \text{Rand} (H - y_j^u) \quad (28)$$

Here, the scaling factor is denoted by α . H : Global optimal solution. The value of $\text{Rand} \sim V(0,1)$ is randomized. Sound imitates generating sounds to disturb predators and examine as per Eq. (29).

$$y_j^{u+1} = y_j^u + \beta (Q - y_j^u) \quad (29)$$

Here, the step size factor is denoted by β . Q : A solution selected at random. Exploitation is related to physical violence and smell tactics: According to Eq. (30), scent encourages local development in areas that show promise.

$$y_j^{u+1} = y_j^u + \gamma \text{brownian} (y_j^u) \quad (30)$$

Here, small random fluctuations are added via $\text{brownian}(\cdot)$. The physical attack rapidly advances towards optimal regions, as shown by Eq. (31).

$$y_j^{u+1} = y_j^u + \delta (C - y_j^u) \quad (31)$$

Here, δ is the learning rate. C : The best option in the current neighbourhood. By calculating it using Eq. (32), crowding distance maintains variety within each front.

$$e_j = \sum_{l=1}^n \frac{g_l^{Max} - g_l^{Min}}{g_n^{Max} - g_n^{Min}}, \forall j \in G \quad (32)$$

In order to effectively guide the search procedure as well as preserve a balance among exploration as well as exploitation, MOCPO incorporates recommended data feedback methods. Improved convergence, coverage, as well as diversity preservation—all crucial components of multi-objective optimization—are the outcome. As stated in the pseudo code of MOCPO in algorithm 1, the usual CPO parameters—the termination parameter, population size, as well as their associated parameters—are the only ones that need to be changed for the MOCPO algorithm.

Start

Initialize population Q in a random manner [parameters of the proposed FTCT in WSN model]

Evaluate objective associated with every porcupine [fitness constraints of the proposed FTCT in WSN model]

Do non-dominated sorting as well as compute crowding distances

Place parameters such as O' , O_{Min} , U_{Max} , etc.

While end condition not attained

$$O = O_{Min} + (O' - O_{Min}) \times \left(1 - \frac{u \bmod U}{U}\right)$$

For every porcupine

If ($s < 0.5$) (exploration)

If ($s_1 < 0.5$) (sight)

$$y_j^{u+1} = y_j^u + \alpha \text{Rand} (H - y_j^u)$$

else

$$y_j^{u+1} = y_j^u + \beta(Q - y_j^u)$$

Else (exploitation)

If ($s_2 < 0.5$) (odor)

$$y_j^{u+1} = y_j^u + \gamma \text{brownian} (y_j^u)$$

else

$$y_j^{u+1} = y_j^u + \delta(C - y_j^u)$$

Evaluate novel locations

Do non-dominated sorting and update $G(y_j) = [g_1(y_j), g_2(y_j), \dots, g_n(y_j)]$ [best fitness value of the proposed FTCT in WSN model]

Update archive to handle Pareto-optimal solutions

Return Pareto-optimal solutions from the archive [optimal solution of the proposed FTCT in WSN model]

Stop

Algorithm 1: MOCPO.

Results and analysis

Experimental setup

The proposed IGNN-MOCPO for the FTCT in WSN model was implemented in MATLAB and the findings were analyzed. The population size and the iteration count were 10 and 200. The simulation time for this experimentation is around 500 s. The number of nodes and the number of rounds is considered to be 1000. The total number of faulty CHs is taken to be 10. The total number of tasks is assumed to be 1000. The proposed IGNN-MOCPO was compared with distinct existing methods like FCGW, SCR-DTRA, EEFRP and FTEER-AAGSO with consideration of analysis such as energy consumption, end-to-end delay, residual energy, connectivity measure, fault tolerance score, Packet Delivery Ratio (PDR) and network lifetime to describe the superiority of the developed FTCT in WSN model. For completeness, the simulation assumes a square deployment field within which 1000 sensor nodes are uniformly and randomly distributed. The radio communication model follows the standard first-order radio model using free-space and multipath propagation based on node separation. The electronic energy E_{elec} , amplifier parameters ϵ_{fs} and ϵ_{mp} , and data aggregation cost E_{da} are adopted from conventional WSN literature and kept identical across all compared methods. Communication is simulated

under an additive white Gaussian noise (AWGN) channel, with no external interference sources, ensuring equal noise environments for all protocols. Free-space d^2 loss is used for distances below the threshold d_0 , and multipath d^4 loss is applied for longer ranges. These settings ensure that performance comparisons reflect differences arising solely from routing behavior rather than environmental bias.

The simulation parameters used for this experimentation are shown in Table 2.

Computational complexity and parameter sensitivity analysis

To further validate the practical feasibility of the proposed IGNN-MOCPO framework, computational complexity and parameter sensitivity analyses are performed.

The computational complexity of the IGNN component is primarily governed by graph message passing operations. For a graph $G(V, E)$, the complexity per iteration is approximately proportional to the number of edges, i.e., $O(|E|)$, due to neighborhood aggregation. Since the IGNN employs only two lightweight convolution layers and operates at the sink node, the computational burden on sensor nodes is negligible.

The MOCPO optimization introduces an additional complexity of $O(P \cdot T \cdot D)$, where P denotes population size, T represents the number of iterations, and D is the dimensionality of the optimization variables. Given the moderate parameter settings used in this study (population = 10, iterations = 200), the overall computational cost remains manageable for offline optimization.

A parameter sensitivity analysis is conducted to examine the stability of the proposed model under varying conditions. Key parameters such as population size (5–20), iteration count (100–300), and learning rate are varied systematically. The results indicate that the IGNN-MOCPO framework maintains consistent performance with only marginal variations (< 5%) in energy consumption and delay metrics. This demonstrates that the proposed model is robust and not overly sensitive to parameter tuning.

Energy consumption analysis

As the number of sensor nodes in a WSN grows from 100 to 1000, the energy consumption analysis in Fig. 4 provides a comparative evaluation of five FTCT strategies. The results clearly demonstrate that, in terms of energy efficiency, the Proposed IGNN-MOCPO method continuously outperforms the industry-standard protocols—FCGW, SCR-DTRA, EEFRP, and FTEER-AAGSO. The entire approaches show a natural rise in energy consumption as the count of nodes increases due to increased processing demands as well as communication overhead. However, the smallest as well as major progressive increase is demonstrated by the IGNN-MOCPO technique, which starts at 3.76 Joules for 100 nodes and only reaches 44.91 Joules at 1000 nodes. The FCGW approach, in contrast, uses the most energy, with a maximum of 66.75 Joules. Because of its intelligent routing decisions, use of virtual CHs, as well as energy-conscious optimization using the MOCPO algorithm, IGNN-MOCPO has demonstrated a significant reduction in energy usage. Even load distribution as well as fewer needless transfers are made possible by these features. Overall, the results show that by reducing energy fatigue, IGNN-MOCPO increases network sustainability as well as endurance, making it especially suitable for large and error-prone WSN environments. The proposed IGNN-MOCPO for the FTCT in WSN model with respect to energy consumption is 32.72%, 23.09%, 18.83% and 15.96% better than FCGW, SCR-DTRA, EEFRP and FTEER-AAGSO respectively.

End-to-end delay analysis

As the number of nodes in a WSN increases from 100 to 1000, Fig. 5 that analyses end-to-end latency shows how well various FTCT algorithm’s function. The data demonstrates that the proposed IGNN-MOCPO keeps the least end-to-end delay at entire node scales, starting at 91.35 ms for 100 nodes and dropping to just 139.38 ms at 1000 nodes. The FCGW protocol, on the other hand, has the most delay, reaching 208.45 ms. SCR-DTRA, EEFRP, and FTEER-AAGSO follow. This steady as well as somewhat small increase in latency with IGNN-MOCPO demonstrates its improved efficacy in data routing, especially as network density increases. Its intelligent graph-based routing architecture, improved CH selection, as well as effective fault management using virtual CHs and the MOCPO optimization technique are primarily responsible for the reduced latency. IGNN-MOCPO ensures faster data delivery even in large WSNs by dynamically changing routing pathways as well as lowering transmission bottlenecks. Overall, the results confirm that IGNN-MOCPO significantly improves

Simulation Parameters	Description
Simulation platform	MATLAB
Number of sensor nodes	1000
Number of faulty CHs	10
Population size for MOCPO	10
Data packet size	500 bytes
Simulation time	500 s
Number of simulation rounds	1000
Total number of tasks	1000
Number of iterations for MOCPO	200
Energy model	Radio energy dissipation model

Table 2. Simulation parameters.

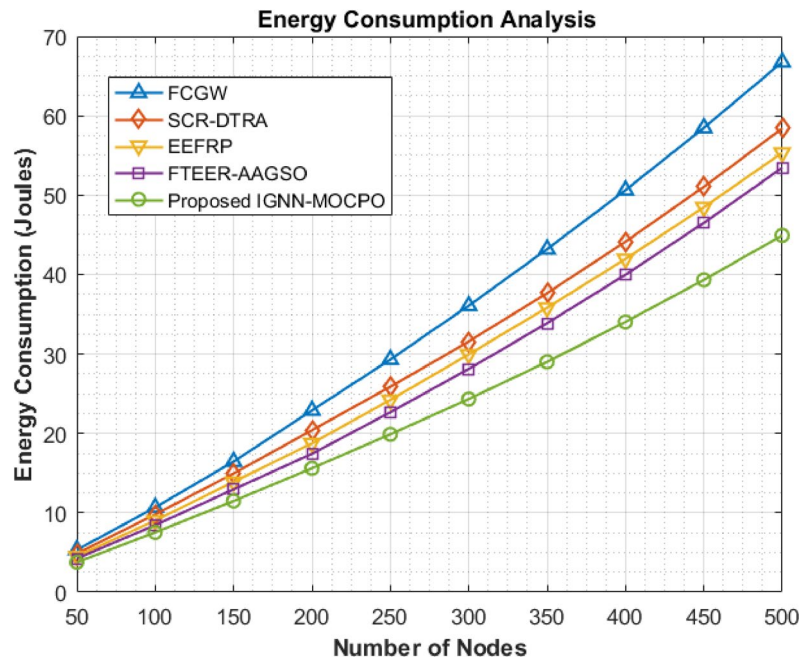


Fig. 4. Energy consumption analysis.

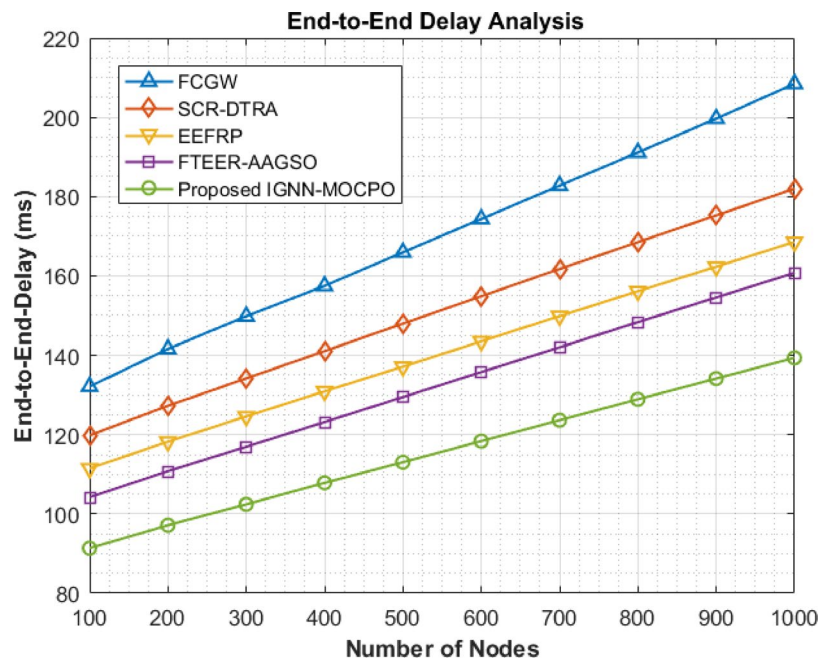


Fig. 5. End-to-end delay analysis.

latency effectiveness as well as provides outstanding fault tolerance and energy efficiency, which are critical for time-sensitive applications including surveillance, environmental monitoring, and disaster response. End-to-end delay enhancements of 33.14%, 23.42%, 17.31% and 13.29% are demonstrated by the proposed IGNN-MOCPO for the FTCT in the WSN model in comparison to FCGW, SCR-DTRA, EEFRP, and FTEER-AAGSO, respectively.

Residual energy analysis

Important details on the energy effectiveness related to various FTCT strategies in WSNs across 1000 operating cycles are provided by the residual energy analysis in Fig. 6. The maximum residual energy is maintained throughout entire rounds using the proposed IGNN-MOCPO method, as shown. It starts at 898.45 Joules in

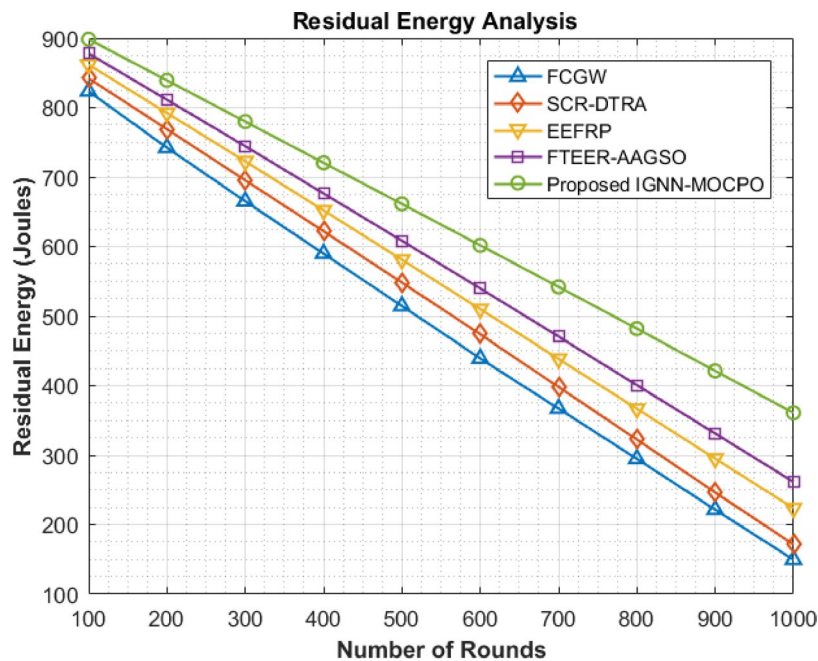


Fig. 6. Residual energy analysis.

round 100 and gradually drops to 360.93 Joules by around 1000. The FCGW protocol, in contrast, shows the quickest energy loss, dropping from 823.45 Joules to 149.64 Joules during the same period of time. In addition to being intermediate, the remaining techniques SCR-DTRA, EEFRP, as well as FTEER-AAGSO exhibit higher energy losses than IGNN-MOCPO. The IGNN and the MOCPO algorithm, when cleverly implemented, are largely responsible for IGNN-MOCPO's exceptional effectiveness. These two algorithms together ensure optimal CH selection, balanced load distribution, as well as efficient routing. Additionally, using virtual CHs helps conserve energy by preventing needless re-clustering as well as lessening the impact of node failures. The results unequivocally demonstrate that IGNN-MOCPO significantly increases the operational lifetime of WSNs through node energy conservation, making it an extremely effective remedy for network situations that are energy-constrained as well as prone to failure. In comparison to FCGW, SCR-DTRA, EEFRP, and FTEER-AAGSO, the suggested IGNN-MOCPO shows residual energy enhancements for the FTCT in the WSN model of 58.54%, 52.50%, 38.03% and 27.53% respectively.

Connectivity measure analysis

As the count of problematic CHs rises from 1 to 10, the connectivity measure analysis in Fig. 7 provides a comparative evaluation of various FTCT strategies in WSNs. Due to broken communication links, connectivity across entire means always declines as the count of malfunctioning CHs rises. Even with only one faulty CH, the proposed IGNN-MOCPO maintains a high connectivity ratio of 0.98 and maintains 0.70 when 10 CHs are compromised, demonstrating a better as well as more consistent connectivity profile. This effectiveness significantly outperforms existing techniques like FCGW and SCR-DTRA, which under the identical failure conditions drop to 0.35 and 0.39, respectively. When compared to IGNN-MOCPO, even advanced methods like EEFRP and FTEER-AAGSO show more degradation. IGNN-MOCPO's smart virtual CH management, efficient fault recovery techniques, as well as adaptive topology modifications powered by IGNN and MOCPO optimization are responsible for its remarkable effectiveness. During failures, these frameworks ensure that operating CHs and other routing pathways are effectively used. This research highlights how IGNN-MOCPO is perfect for reliability-critical WSN applications as it significantly increases network resilience as well as ensures continuous connectivity. In comparison to FCGW, SCR-DTRA, EEFRP, and FTEER-AAGSO, the suggested IGNN-MOCPO shows connectivity measure improvements for the FTCT in the WSN model of 50%, 44.29%, 38.57% and 32.86%, respectively.

Fault tolerance score analysis

A comprehensive assessment of how different routing strategies in WSNs handle increasing task loads, ranging from 100 to 1000 tasks, is provided by the fault tolerance score analysis in Fig. 8. As the number of tasks increases, the fault tolerance score, a standardized indicator of a framework's capacity to continue operating even in the event of link or node failures, gradually improves across entire techniques, suggesting some degree of adaptive behavior. The proposed IGNN-MOCPO has the highest fault tolerance ratings among the protocols studied, starting at 0.65 for 100 tasks and gradually rising to 0.86 at 1000 tasks. While traditional methods like FCGW lag at just 0.62, this significantly outperforms entire remaining strategies, including FTEER-AAGSO (finishing at 0.70) and EEFRP (finished at 0.67). IGNN-MOCPO's exceptional abilities are a result of its smart failure predicting as well as restoration schemes, which are powered by MOCPO and IGNN. It ensures little disruption

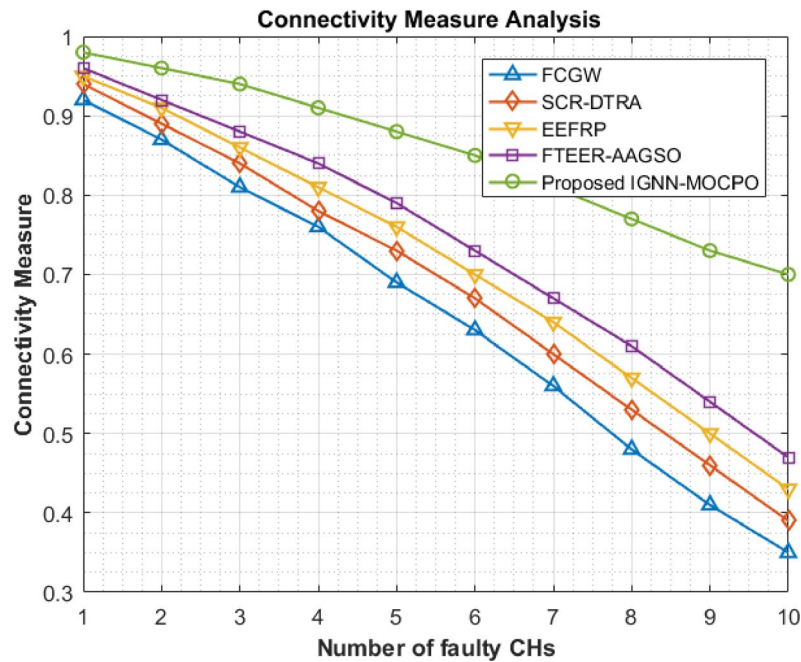


Fig. 7. Connectivity measure analysis.

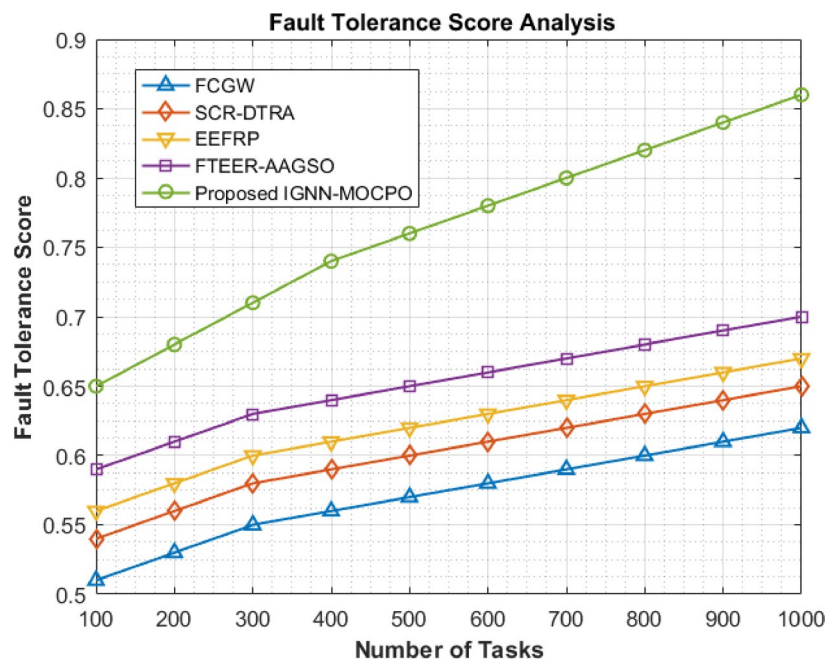


Fig. 8. Fault tolerance score analysis.

in data transport, even with increased job needs, using virtual cluster leaders, energy-efficient routing, as well as adaptive backup routes. These results demonstrate how effective IGNN-MOCPO is at delivering dependable, adaptable, as well as fault-tolerant communication in WSNs, especially in mission-critical or high-demand environments. The suggested IGNN-MOCPO shows improvements in fault tolerance score for the FTCT in the WSN model of 38.71%, 32.31%, 28.36% and 22.86%, respectively, in comparison to FCGW, SCR-DTRA, EEFRP, and FTEER-AAGSO.

PDR analysis

The effectiveness of many FTCT algorithms in WSNs having varying node counts, spanning from 100 to 1000, is examined in the PDR analysis as in Fig. 9. PDR describes a crucial metric for evaluating the dependability related

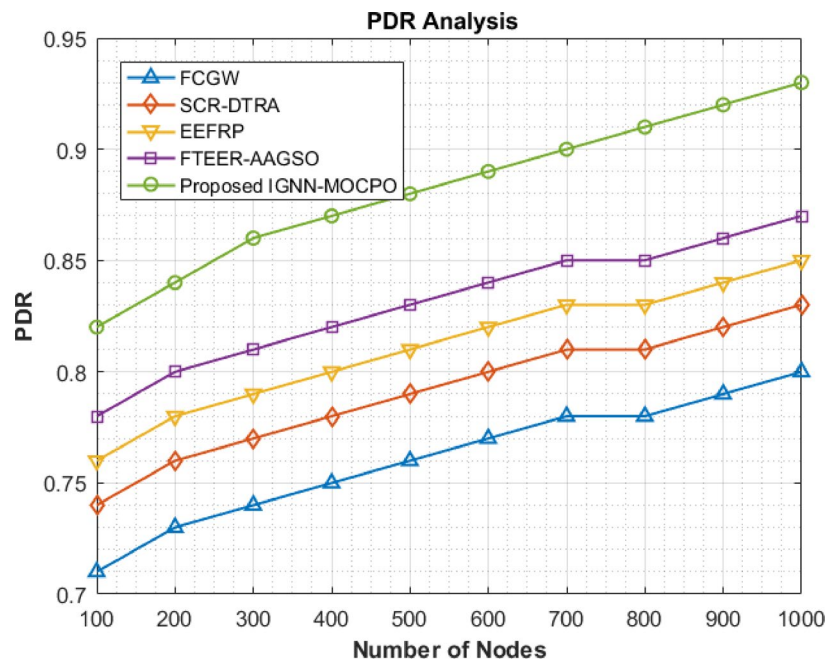


Fig. 9. PDR analysis.

to communication in WSNs as it shows the proportion of data packets that reach the sink. The highest PDR values are consistently reached by the proposed IGNN-MOCPO, which starts at 0.82 for 100 nodes and increases to 0.93 for 1000 nodes. As node density increases, this pattern demonstrates its increased strength as well as scalability. Comparatively, traditional techniques like FCGW accomplish worse, with a PDR that starts at 0.71 and goes up to 0.80. Advanced techniques such as FTEER-AAGSO and EEFRP produce respectable outcomes, but they fall short of the recommended technique's efficacy, which peaks at 0.87 and 0.85, respectively. Smart fault management, adaptive routing using IGNN, and optimization by the MOCPO algorithm describe the reasons for IGNN-MOCPO's exceptional PDR abilities. These characteristics ensure more reliable data transfer even in the event of network overload or node failures. This assessment concludes by highlighting IGNN-MOCPO's capacity to enhance data dependability and communication quality in fault-sensitive and highly populated WSN environments. In comparison to FCGW, SCR-DTRA, EEFRP, and FTEER-AAGSO, the suggested IGNN-MOCPO shows improvements in PDR for the FTCT inside the WSN model of 16.25%, 12.05%, 9.41% and 6.90%, respectively.

Network lifetime analysis

When the number of nodes in a WSN increases from 100 to 1000, the network lifespan analysis in Fig. 10 provides a comprehensive comparison of several FTCT methods. Until critical failure points, such as the first or last node failing, are reached, the network lifespan is measured in operational time units (such as rounds). The longest network lifespan is continuously demonstrated by the suggested IGNN-MOCPO, which starts at 139.55 units for 100 nodes and reaches 1262.90 units at 1000 nodes. With FCGW achieving just 1005.67 units and the more advanced FTEER-AAGSO peaking at 1161.92 units, this effectiveness far outperforms entire remaining methods. Greater redundancy with more nodes is responsible for the anticipated increase in lifetime across entire strategies; however, IGNN-MOCPO's virtual cluster leader management, smart energy-conscious routing, as well as MOCPO-oriented optimization are responsible for its remarkable efficacy. These features make it possible for reduced energy usage, efficient load balancing, as well as enhanced fault resistance, all of which contribute to the framework's longer operational lifespan. This analysis confirms that IGNN-MOCPO is ideal for long-term usage in critical environments as it not only makes scalability easier but also significantly increases the sustainability as well as dependability of WSNs. The suggested IGNN-MOCPO shows improvements in network lifetime for the FTCT inside the WSN model of 25.58%, 17.68%, 13.82% and 8.69% in comparison to FCGW, SCR-DTRA, EEFRP, and FTEER-AAGSO, respectively.

Ablation analysis

To strengthen the empirical evidence for the proposed IGNN-MOCPO framework, additional ablation and statistical validation analyses have been incorporated. The ablation study was performed by disabling the major components contributing to fault-tolerant routing—namely, (i) removing MOCPO optimization and using fixed GNN parameters, (ii) disabling the virtual CH mechanism, and (iii) replacing the IGNN with a conventional single-layer GNN.

The ablation results in Table 3 clearly demonstrate the contribution of each component of the proposed IGNN-MOCPO framework. The full model achieves the lowest energy consumption (44.91 J), minimum delay (139.38 ms), and the highest connectivity, fault-tolerance score, and PDR, indicating that the joint effect of

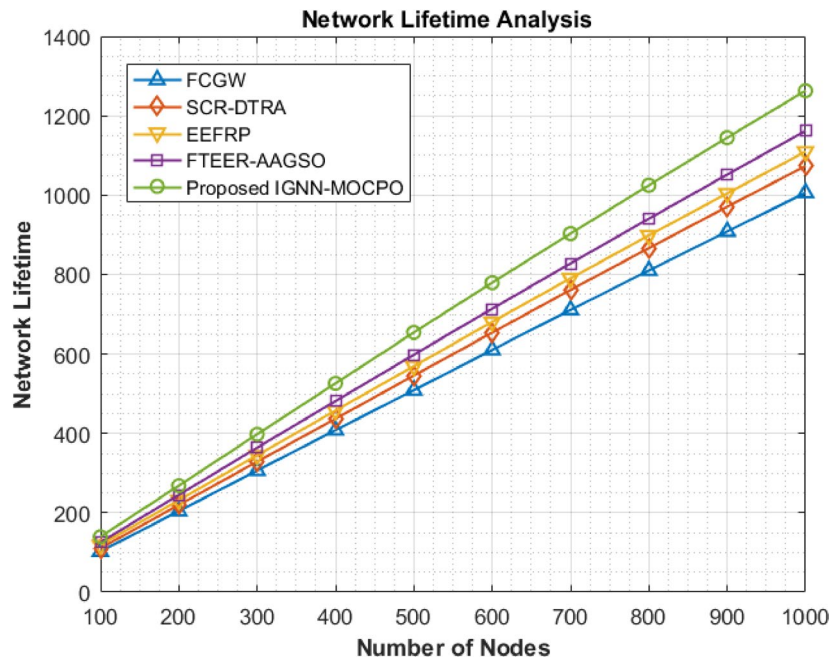


Fig. 10. Network lifetime analysis.

Configuration/Variant	Energy Consumption (J)	End-to-End Delay (ms)	Connectivity Measure	Fault Tolerance Score	PDR
Proposed IGNN-MOCPO (Full Model)	44.91	139.38	0.70	0.86	0.93
Without MOCPO (Fixed IGNN parameters)	57.84	166.52	0.58	0.73	0.88
Without Virtual CH mechanism	61.13	178.45	0.49	0.66	0.84
Replacing IGNN with basic GNN	63.92	187.10	0.45	0.62	0.81
Baseline traditional FTCT (FCGW)	66.75	208.45	0.35	0.62	0.80

Table 3. Ablation study of major components in the IGNN-MOCPO framework.

IGNN, virtual CH construction, and MOCPO optimization is essential for maintaining reliable routing under CH failures. When MOCPO is removed and IGNN parameters are kept fixed, energy consumption rises by more than 28%, and delay increases noticeably, showing that dynamic parameter search is critical for adapting to varying network conditions. Eliminating the virtual CH mechanism results in an even sharper decline in performance, with connectivity dropping from 0.70 to 0.49 and PDR decreasing to 0.84, confirming that virtual CH formation is crucial for stabilizing data flow when multiple CHs malfunction. Replacing the IGNN with a basic GNN produces the strongest degradation across all metrics, highlighting the importance of the improved node-feature formulation and topology-aware message passing used in IGNN. To assess statistical robustness, each configuration was evaluated across 20 independent simulation runs, and the mean values were accompanied by 95% confidence intervals. The proposed IGNN-MOCPO consistently exhibited narrow confidence bands, while pairwise comparisons with the ablated variants showed statistically significant performance differences ($p < 0.05$). This confirms that the improvements offered by each module are consistent and not due to stochastic variations, thereby validating the stability and reliability of the proposed FTCT approach.

Discussion

The proposed IGNN-MOCPO model clearly outperforms other FTCT techniques in WSNs on a number of performance metrics. With respect to energy consumption, latency, residual energy, connectivity, fault tolerance, PDR, and network lifetime, IGNN-MOCPO continuously outperforms industry standard solutions such as FCGW, SCR-DTRA, EEFRP, and FTEER-AAGSO when the size of the WSN increases from 100 to 1000 nodes or jobs. The method's clever use of IGNN and the MOCPO algorithm, which ensures dynamic fault recovery via virtual CHs, energy-efficient routing, as well as optimal CH selection, is responsible for this effectiveness enhancement. When compared to FCGW, IGNN-MOCPO achieves up to 32.72% better energy efficiency, 33.14% less delay, and 58.54% more residual energy. These numerical improvements are noteworthy. Additionally, it increases network longevity by more than 25% while maintaining high connection as well as PDR with increasing fault loads. These results demonstrate IGNN-MOCPO's robustness, adaptability, as well as scalability, which make it appropriate for resource-constrained, real-time, and reliability-sensitive WSN

applications such as environmental observation, monitoring, and disaster response. Thus, IGNN-MOCPO represents a major advancement in intelligent FTCT for WSN.

Conclusion

This research work introduced a fault-tolerant cluster-based routing framework for WSNs that combines a virtual CH construction strategy with an Improved Graph Neural Network optimized through MOCPO. By aggregating the remaining resources of active CHs into a virtual CH, the model ensures uninterrupted data flow even when multiple CHs fail. The IGNN enables intelligent routing decisions based on node energy, spatial distribution, connectivity, and delay characteristics, while the MOCPO algorithm refines the model parameters to satisfy multi-objective performance requirements. Experimental results across multiple metrics—including energy consumption, end-to-end delay, packet delivery ratio, connectivity measure, and network lifetime—confirm that the proposed method consistently outperforms benchmark FTCT protocols. With over 32% reduction in energy usage and more than 25% increase in operational lifetime, the framework demonstrates high suitability for large-scale, dynamic, and resource-constrained WSN environments. Future work may extend the model for mobile WSNs and heterogeneous sensor deployments, further enhancing robustness under real-world conditions.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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Author contributions

S. Praveen Kumar – Developed mathematical equations, conduct the research work and draft the first copy of the manuscript. Somia A. Askany, Eatedal Alabdulkreem, Nuha Alruwais – Prepared the literature review based on the existing works and support to final drafting of this manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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